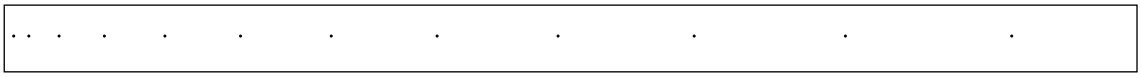
Ticker tapes and oil drops

12a

A thin paper tape passing through a buzzer was attached to a mechanical mouse. The idea was to use the buzzer and its imprint on the tape to make a record of the mouse's speed over a period of time. The buzzer makes an impact on the tape once every tenth of a second; so ten impacts on the tape mark the passing of one second. Below is the beginning of one of these tapes.



1 Describe the trend about the distance between dots as the mouse moved along.

_____

2 Number the gaps between two adjacent dots (staring at the left of the tape with the smallest gap being numbered '1').

3 Measure the distance between every two dots and record the gap distances in the table below.

Gap number	1	2	3	4	5	6	7	8	9	10	11
Gap distance (mm)	2										
Gap time (s)	0.1										
Average speed (mm/s)	20										

4 Complete the row about gap times.

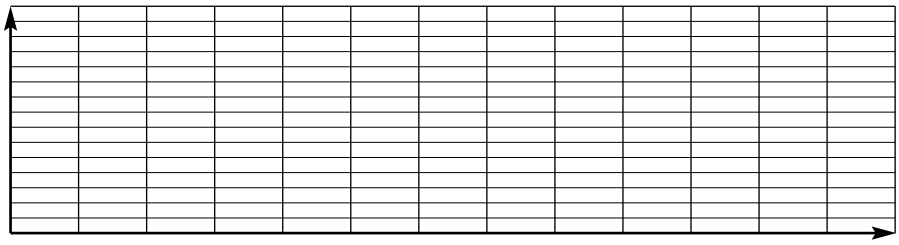
5 State the formula for calculating speed. _____

6 Calculate the average speed for every gap and write it in the above table.

7 Describe the trend about the speed as the mouse moves along.

_____

8 **a** Cut across the tape at every dot. Glue the pieces in order from smallest to largest onto the graph below. Label the axes of the graph appropriately, and give it a title.



b Connect the dots at the top of the tape pieces and draw a smooth line through them.

c State what the smooth line represents.

_____

d Calculate the overall average speed from your graph.

_____

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12b

9 Acceleration can be found by subtracting the initial speed from the final speed, and dividing the answer by the time between the speed readings. Write the formula for calculating acceleration.

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10 **a** Calculate the acceleration at three different places in your graph on the previous page. Use the table below as guide.

Initial speed (mm/s)	Final speed (mm/s)	Speed difference (mm/s)	Time for speed difference (s)	Acceleration (mm/s/s)

b Describe the trend about the acceleration as the mouse moves along.

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11 A car travelling along a road has been dropping oil, one drop of oil every second. Part of the oil trace is shown on the following scaled diagram.

(Key: 1 mm on the drawing = 1 m on the road.)

Start Finish

. . . . . .

a Indicate between successive drops whether the car was accelerating (ac), decelerating (de) or travelling at a constant speed (co). Mark these symbols on the diagram above.

b Calculate the car's speed at six different places along the oil trail. Number your chosen places 1, 2, etc. Use the table to record distances, calculations and speeds.

Interval	Distance (drawing)	Distance on road	Distance/time	Average speed
1	3 mm	3 m	3/1 ms^{-1}	3 m/s
2				
3				
4				
5				
6				

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